

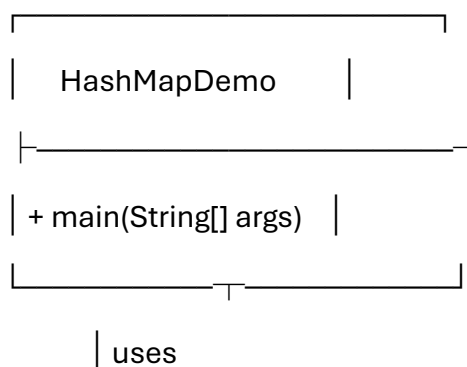
Aim

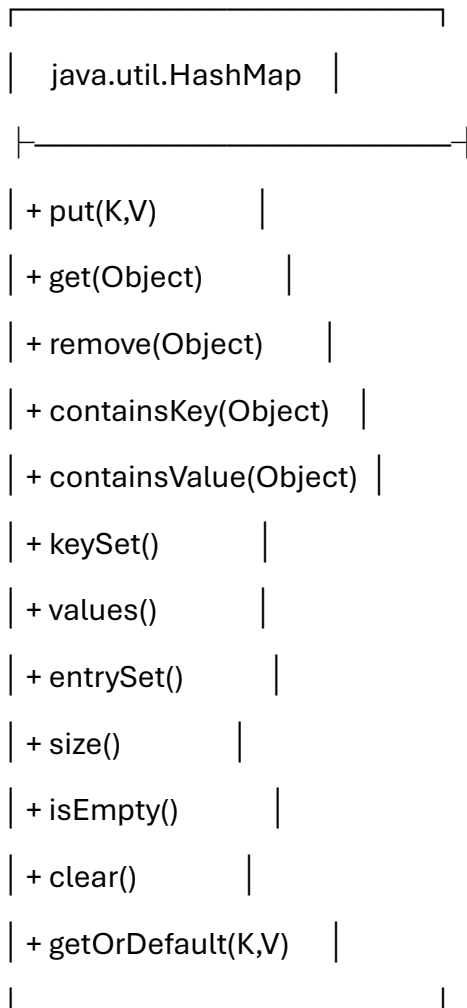
To write a Java program to demonstrate the important methods of the **HashMap class**, such as put(), get(), remove(), containsKey(), containsValue(), keySet(), values(), entrySet(), size(), isEmpty(), clear(), and getOrDefault().

Algorithm

1. Start the program.
2. Import the java.util.HashMap class.
3. Create a HashMap object.
4. Add key-value pairs using put().
5. Retrieve a value using get().
6. Check for a key and value using containsKey() and containsValue().
7. Remove a mapping using remove().
8. Display keys using keySet().
9. Display values using values().
10. Display key-value pairs using entrySet().
11. Find the number of mappings using size().
12. Check whether the map is empty using isEmpty().
13. Use getOrDefault() to retrieve a value or a default value.
14. Remove all mappings using clear().
15. Display the results.
16. Stop the program.

UML Diagram





Code:

```
package Observations;
```

```
import java.util.HashMap;
```

```
import java.util.Map;
```

```
public class HashMapClass {
```

```
    public static void main(String[] args) {
```

```
        // Creating HashMap
```

```
        HashMap<Integer, String> map = new HashMap<>();
```

```
// put()

map.put(101, "Apple");
map.put(102, "Banana");
map.put(103, "Mango");
map.put(104, "Orange");

System.out.println("HashMap: " + map);

// get()

System.out.println("Value for key 102: " + map.get(102));

// containsKey()

System.out.println("Contains key 103: "
    + map.containsKey(103));

// containsValue()

System.out.println("Contains value Mango: "
    + map.containsValue("Mango"));

// getOrDefault()

System.out.println("Value for key 105: "
    + map.getOrDefault(105, "Not Found"));

// remove()

map.remove(104);

System.out.println("After removing key 104: " + map);

// keySet()
```

```
System.out.println("Keys: " + map.keySet());

// values()
System.out.println("Values: " + map.values());

// entrySet()
System.out.println("Key-Value pairs: " + map.entrySet());

// size()
System.out.println("Size of HashMap: " + map.size());

// isEmpty()
System.out.println("Is HashMap empty? " + map.isEmpty());

// clear()
map.clear();
System.out.println("After clear(): " + map);

System.out.println("Is HashMap empty after clear()? "
    + map.isEmpty());
}
```

Output:

```

es' '-cp' 'C:\Users\Sahithi_pyla\AppData\Roaming\Code\User\workspaceStorage\2340
bservations.HashMapClass'
HashMap: {101=Apple, 102=Banana, 103=Mango, 104=Orange}
Value for key 102: Banana
Contains key 103: true
Contains value Mango: true
Value for key 105: Not Found
After removing key 104: {101=Apple, 102=Banana, 103=Mango}
Keys: [101, 102, 103]
Values: [Apple, Banana, Mango]
Key-Value pairs: [101=Apple, 102=Banana, 103=Mango]
Size of HashMap: 3
Is HashMap empty? false
After clear(): {}
Is HashMap empty after clear()? true
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>

```

Result:

Thus, the Java program successfully demonstrates the important methods of the HashMap class for storing and manipulating key-value pairs.